Isabella (Yaoyuan) Hu

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Portfolio: https://ihu3025.github.io/IHU.github.io/| **Github**: https://github.com/IHU3025
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Education

University of California, Berkeley | Berkeley, CA | **GPA**: 3.93 Fall 2023 - 2027
Bachelor of Arts in Computer Science and Applied Mathematics
Relevant Coursework: Programming Structure, Data Structures, Data Management, Discrete Mathematics, Abstract Linear Algebra, Abstract Algebra, Probability, Computer Graphics, Computer Vision, Machine Learning

Skills

Programming Languages: Python, Java, C#, C++, JavaScript, ReactJS, HTML&CSS
Technical Tools: Unreal Engine, Unity, OpenGL, Matlab, Git, Docker,
Animation & VFX: Houdini, Maya, Blender, ZBrush, After Effects, Adobe Premiere, Photoshop

Internship Experience

AI Department Intern | Vancouver Film School **Dec 2024 - Jan 2025**

- Engineered an AI pipeline integrating **NVIDIA Riva** and **Audio2Face** with **Unreal Engine** to drive real-time facial animation for an **AI-Powered Digital Human** interactive installation.

AI Innovation Research Intern | Shanghai Film Group Co. Ltd. (SFG) **June – July 2024**

- Evaluated a portfolio of 50+ AI tools for viability across production workflows and authored the AI in Animation & Film section of an internal strategic report to guide company-wide R&D and tool adoption.

Work Experience

Compsci 61A Tutor | UCB EECS Department **Sept - Dec 2024, May 2025 - Present**

- Lead 2 weekly exam-prep sessions for 10 students; developed and delivered mini-lectures and practice problems to strengthen conceptual understanding and exam strategy
- Hosted 5+ weekly office hours, providing 1-on-1 support to students to debug code and clarify concepts in **Python, Scheme, and SQL**

Research

Rehabilitation with ScenicXR Research **Feb - May 2025**
Deployed webportal: **https://reia-rehab.com.**

- Built and deployed **full-stack features** for the Reia Rehab portal, developing the **ReactJS** front-end and **next.js** back-end for modules including the *Patient List Page*, *Exercise Prescription* page and the *Scheduler*
- Integrated OpenAI transcription for **AI-driven VR exercise generation**, streamlining data processing.
- Coordinated **cloud-based solutions** for data storage and secure access with the backend team.
- Engineered a real-time remote patient-monitoring prototype using **MediaPipe** to classify exercise repetitions.

Technical Projects

Isotropic Remeshing Pipeline for Physical Simulation | C++, Eigen Lib , OpenGL **Aug 2025**

- Engineered an **isotropic remeshing pipeline** to eliminate mesh degeneracy in a hyperbolic flow simulation

https://github.com/CarlSXH/Bubble | https://carlsxh.github.io/CS184/finalproject/deliverable/index.html

Physically-Based Path Tracer with Global Illumination | C++, GLSL, OpenGL **July 2025**

https://lapluma430.github.io/cs184hw_website/hw3/index.html

VR Prototype: Halloween Environment Game | Unity, C# | Personal Project **Sept 2024**

- Developed an immersive **VR experience in Unity**, implementing **locomotion**, object interaction, and **enemy AI** behavior systems scripted in **C#**.

https://github.com/IHU3025/HalloweenVR